

Second Edition

RESEARCH METHODOLOGY IN APPLIED ECONOMICS

Organizing, Planning, and Conducting
Economic Research

Don Ethridge

 **Blackwell**
Publishing

Don E. Ethridge, Ph.D., is Chairman and Professor, Department of Agricultural and Applied Economics, Texas Tech University, Lubbock.

©2004 Blackwell Publishing
All rights reserved

Blackwell Publishing Professional
2121 State Avenue, Ames, Iowa 50014, USA

Orders: 1-800-862-6657
Office: 1-515-292-0140
Fax: 1-515-292-3348
Web site: www.blackwellprofessional.com

Blackwell Publishing Ltd
9600 Garsington Road, Oxford OX4 2DQ, UK
Tel.: +44 (0)1865 776868

Blackwell Publishing Asia
550 Swanston Street, Carlton, Victoria 3053, Australia
Tel.: +61 (0)3 8359 1011

Authorization to photocopy items for internal or personal use, or the internal or personal use of specific clients, is granted by Blackwell Publishing, provided that the base fee of \$.10 per copy is paid directly to the Copyright Clearance Center, 222 Rosewood Drive, Danvers, MA 01923. For those organizations that have been granted a photocopy license by CCC, a separate system of payments has been arranged. The fee code for users of the Transactional Reporting Service is 0-8138-2994-1/2004 \$.10.

Printed on acid-free paper in the United States of America

First edition, © 1995, Iowa State University Press
Second edition, © 2004

Library of Congress Cataloging-in-Publication Data

Ethridge, Don E. (Don Erwin)

Research methodology in applied economics: organizing, planning, and conducting economic research/Don Ethridge—2nd ed.

p. cm.

Includes bibliographical references and index.

ISBN 0-8138-2994-1 (alk. paper)

1. Economics—Research. 2. Economics—Methodology. I. Title.

HB74.5.E84 2004

330'.9072—dc22

2004001461

The last digit is the print number: 9 8 7 6 5 4 3 2 1

The Conceptual Framework

Economics is a science of thinking in terms of models joined to the art of choosing models which are relevant to the contemporary world. Good economists are scarce because the gift of using "vigilant observation" to choose good models, although it does not require a highly specialized intellectual technique, appears to be a very rare one.

John Maynard Keynes

As noted previously, a formal conceptual framework is not required in all research proposals. Whether required or not, formal or not, no economic research endeavor should be undertaken without benefit of the researcher(s) having developed a conceptual framework for the project. The objective of this chapter is to define, explain, and provide guidelines for developing the conceptual framework within a research proposal or project. In this chapter it is assumed that a formal, written conceptual framework is a part of the written research proposal.

There may be more misunderstanding and confusion about the conceptual framework than any other part of the research process (or research proposal). Confusion exists on what the conceptual framework is, what its purposes are, and how it is approached. Many economists fail to understand it clearly, so there is little wonder that students of economics are confused. Students typically resist undertaking the writing of a formal conceptual framework unless given no other alternative. Developing the conceptual framework is an integral part of understanding and analyzing the research problem rather than some vague, abstract addendum to the research. The conceptual framework should be understood as an integral component of framing the research problem. The successful development of a conceptual framework helps one understand the problem. In many cases, a clear statement of the problem does not occur until the conceptual framework is developed. It is complementary to the problem description, statement of objectives, and literature review, instead of the fourth step in a series of steps. We treat

it as a distinct piece of the proposal in order to provide organization to the proposal, but the reasoning presented in it is intertwined with the previous sections of the proposal.

The conceptual framework is often neglected or hastily formulated, at least in a relative sense, in the research process. One reason for this neglect may be because it requires a somewhat complex set of abilities and characteristics—abstract reasoning, recognition and synthesis of central points, knowledge of existing economic theory, motivation to understand the details of a problem and the surrounding issues. Conceptualizing is also hard work and can be frustrating. Another factor that contributes to the neglect, especially in subject-matter and problem-solving research, is that we may want to get on with the “real research” instead of “theorizing.” This attitude is particularly prominent in students, who do not realize that the conceptualizing is an integral part of the research and that failure to conceptualize while *planning* the research may act to slow the research, if not cause fatal errors.

An example may help to illustrate both the importance of developing the conceptual framework and the hazards of not doing it rigorously. Following the U.S. government's breakup of American Telephone and Telegraph (AT&T) in the early 1980s, an editorial appeared in *Newsweek* magazine. The editorial noted that the dissolution of the AT&T monopoly had led to lower long-distance rates but higher local telephone service rates, contrary to the expectation that lower local rates would result from the breakup of the monopoly (Rosenberg, 1983). The writer was correct in his analysis of long-distance rates—the increased competition from other long-distance companies resulted in lower rates. However, his conceptual analysis of the local service rates was erroneous. He failed to recognize that the breakup of the one national company for local service into seven regional companies did not dissolve the monopoly; it replaced the one national monopoly with seven regional monopolies. He apparently assumed, without basis, that size and monopoly power are synonymous. The primary point is that one cannot be assured of a full understanding of the problem without a rigorous conceptual framework.

Previous writings on the conceptual framework tend to be piecemeal and sparse in comparison with writings on the other components of the research proposal. In this chapter, unpublished course notes of Willard Williams (1984) and discussions, debates, and arguments with him on the matter as a student and a colleague are drawn upon heavily. The author's understanding and perspective of the conceptual framework, its purposes, and its value have been influenced more by his thinking on the topic than by any other single individual.

The chapter is organized as follows. The first section defines the conceptual framework. The second section addresses the use of economic theory in the conceptual framework. The third section addresses hypotheses and hypothesis testing, and the fourth offers suggestions on formulating and writing the conceptual framework.

ROLE OF THE CONCEPTUAL FRAMEWORK

Webster's New Collegiate Dictionary (1977) defines *concept* as "something conceived in the mind" and *conceptual* as "of, relating to, or consisting of concepts." Lacey (1976), in his *Dictionary of Philosophy*, explains that to have a concept of anything is to be able to think or reason about it; for example, to have a concept of dog is to be able to think about dogs. Concepts are generalizations.

For use within the confines of economics and economic research, we can define *concept* as *a logical, mental construction of one or more relationships*. Several attributes of this definition are important. It is purely mental, is logical, and can be described; it has been reasoned through sufficiently and presented with clarity. As such, a concept is inherently abstract (takes some things as given or assumed). In economics, concepts typically focus on relationships; they may be basic relationships among variables or more complex systems of relationships. Also, the relationships are causal in nature. That is, the relationships intend to explain how or why some things result in (cause) other things.

In economic research the conceptual framework is a conceptual analysis through the problem to all hypotheses relevant to the problem. (Williams, 1984)

Note that the conceptualization is directed toward the problem, not the objectives or the methods. It is purely conceptual, that is, without regard for empirical evidence or data. Its primary function is to lead to and justify meaningful hypotheses that are, in turn, subject to testing (verification or rejection).

Most of us conceptualize as a part of our everyday living, although we may not be aware of it, and therefore the process is probably not analytically sophisticated. The following illustrates the point:

Who has not entered his automobile only to find that it will not start? With presentation of the problem, the conceptualizing process begins. The driver begins to reason concerning causes and possible solutions. Since the starter works and the engine turns over, the problem is neither the battery nor the starter. He might continue reasoning in this manner and, in the process, narrow both the possible causes and alternative solutions. (Williams, 1984)

Grasping the purpose of the conceptual framework is fundamental to using it effectively. The conceptual framework is not a test of the researcher's knowledge of existing economic theory. In disciplinary research, it could entail the development or new theory, however. It is not the mathematical derivation of a research method that is planned to be used in a study unless the intent of the research is to develop a new method. It is not the explanation of the logic of an econometric model. It is not the presentation of the standard economic theory that may be applicable to a given study. Yet the conceptual framework may be, and often is, related to these other activities. The relatedness to methods and objectives is,

however, through the side effects of the conceptual framework, not as its main purpose or its focus of attention. Diversion of attention to the side effects rather than its function can lead one to overlook or misunderstand important dimensions of the research problem.

The conceptual framework may be viewed as an analysis of the research problem(s) using theory. The theory will probably include economic theory if the research problem is defined by an economist, but it may not necessarily exclude other theory. Theoretical considerations from other disciplines may be appropriate to the conceptual economic analysis. It is most often a process of identifying the appropriate economic and other theory or concepts that are germane to the analysis of the problem, then applying them in a conceptual analysis of that specific problem. Examine the conceptual frameworks in the proposals in appendices A and C. Both address the specific problem identified in the research proposal. That is, the central focus of the conceptual analysis is the main problem issue identified in the specific problem statement. Yet in the case of the proposal in appendix A there is a direct lead-in to the methods and procedures.

If the theory is not already developed, the task may involve formulating theory or refining or modifying existing theory. Theoretical formulations or refinements may be more likely to be required when the research is more oriented to disciplinary matters, as opposed to a heavier emphasis on problem-solving or subject-matter research. This generalization is, however, quite fallible. It is not uncommon for applied research to identify the need for theoretical refinements and result in refinements being developed within the conceptual framework. When that occurs, the research takes on more prominent disciplinary elements.

Although problem-oriented, the conceptual framework often results in other benefits (the side effects mentioned above). One is that it may provide a theoretical link between the objectives and the methods and procedures. The conceptual framework is related to the objectives because the problem leads directly to the objectives. The conceptual analysis thus may help identify relationships, or types of relationships, that are needed to achieve the objectives. It is common for the conceptual framework to point to relevant variables within relationships as well. Williams (1984) characterizes the conceptual framework as an organized "think piece" that, in its analysis of the problem, may include the logic of:

- A. Sources of the problem. This may address conditions, circumstances, policies, practices, etc., that cause(d) the problem.
- B. Alternative solutions to the problem.
- C. Identification of variables relevant to the analysis of the problem.
- D. Conceptualized relationships in a system to analyze the problem.
- E. Hypotheses to be tested about results of analysis on the problem.

SOURCE MATERIAL FOR THE CONCEPTUAL FRAMEWORK: THEORY

How does one get started on the conceptual framework? Look to the relevant economic and other theories. Which theory is relevant? The relevant theory provides insight into or understanding of the logic of one or more of the five elements just identified, while focusing on the specific researchable problem of the study. This is why a grasp of the theories within a discipline is essential to the conduct of subject-matter and problem-solving research as well as disciplinary research. Placing the problem into a researchable context necessitates its conceptualization which, in turn, requires one to be capable of using the discipline's theoretical tools as a means of *thinking about the problem*.

Recognizing the relevant theory is only the initial step. The researcher is then faced with *applying* that theory to the specific problem. Ladd (1991) describes this as finding the linkages between the "ideal types" in theory and the "real types" that exist in actuality. If the theory relevant to the problem is well developed in its general form, it will invariably require some adaptation to "fit" the individual research problem. When we conceptualize, we select general ideal types from the body of theory and create our own ideal types within the specific research activity. But when conceptualizing involves only the adaptation of existing theory to a research problem, it is still demanding and requires original thinking. Each research problem is unique, and there are no easy guidelines for conceptualizing.

Consider that the conceptual framework in appendix A adapts standard Marshallian welfare concepts of producer and consumer surplus and international trade concepts to the specific problem of understanding the social costs and benefits resulting from a defined set of domestic commodity and trade policies. In appendix C, the conceptual framework uses elements of merger and investment theory in its conceptual analysis of the expected impacts of mergers in pharmaceuticals on the various stakeholder groups. The conceptual framework in appendix B examines the linkages, flows, and types of impacts caused by the incidence of HIV/AIDS in Russia's macroeconomy. Each of these appendix examples employs a different set of theories as a point of departure, yet each adapts well-understood and documented concepts to its research problem, giving clarity to the understanding of the problem.

For another example, if one is concerned with the subject-matter research problem of water use efficiency of the Nile River in Egypt, it is probable that the general theory of efficiency in the use of limited but replenishable natural resources is relevant to the conceptual analysis of the problem. The theory would be adapted to a particular natural resource (water, and specifically the water of the Nile) in a particular place (Egypt or its subdivisions) in its various uses (agriculture, domestic, industrial). That is, we are not interested in restating the general concepts of resource use efficiency; we can read about those principles elsewhere.

We are interested in understanding the dimensions of the problem of allocation and use efficiency in Egypt. Application and adaptation of the general theory to that situation can enhance our grasp of the problem. The conceptual analysis of the researchable problem may give insight into how to achieve the objectives of the study as well.

Following the same example, it is also likely that Egyptian policies will need to be considered in conceptualizing the nature of the problem. That is, one may need to superimpose water policy conditions or constraints on the theoretical construction of water use efficiency to see how one would expect those conditions to affect efficiency. Egypt's policy of free water for agricultural use affects water use efficiency and may need to be examined or explained conceptually. This also would be part of the conceptual framework.

Sometimes adaptation of existing theory is not sufficient for the analysis of the research problem. In these cases one is left with the choice of either refining the theory or presenting an incomplete conceptual framework. The latter option is obviously less than ideal from a disciplinary perspective. Without advocating this option, it is realistic to recognize that identifying the gap in our conceptual understanding is better than ignoring the problem altogether; that is, the conceptual framework that does not develop the theory makes some contribution by identifying the need for more theoretical development. This alternative, while not the most desirable, may be defensible in some situations—in, say, master's theses—but is less likely to be acceptable in Ph.D. dissertations.

When theoretical refinements are introduced, they are primarily as marginal improvements in the existing theory or adaptation of concepts from other subject-matter areas to a new one. "New theory" rarely, if ever, springs forth in full bloom from the sudden insight of an individual. It is too intricate and logically demanding to occur that way. Instead, it evolves through a process of individuals making sequential, marginal contributions to the broader body of knowledge. Even the "breakthroughs" in economic theory—for example, Alfred Marshall's *Principles of Economics* and John Maynard Keynes's *General Theory*—drew heavily on prior theoretical development.

HYPOTHESES AND HYPOTHESIS TESTING

Recall that a primary function of the conceptual framework is to lead to hypotheses that are relevant to the research project. Hypotheses are the results of the reasoning process embodied in the conceptual framework. They need to be formulated in such a way that they are nontrivial, testable, and capable of being refuted (Ghebremedhin and Tweeten, 1994, p. 47). Recall also that we described a *hypothesis* in chapter 3 as *a tentative assertion that is subject to testing*. As a tentative assertion, it can take the form of a simple proposition of an expected outcome or an assertion of a relationship, or relationships, between or among forces,

variables, or events. For example, a simple proposition might be that one production system in a manufacturing operation, based on a particular technology, is more profitable than another production system based on another technology (hypotheses consisting of simple propositions are not limited to problem-solving research projects). An example of a hypothesis of a relationship in, say, a demand study on wine might be that per capita wine consumption in the United States is affected by price of wine, prices of a set of other beverages, per capita income, consumers' religious affiliations, and ethnic background. (See chapter 3 to review the discussion of hypotheses.)

Hypotheses are not always stated explicitly, especially in economic research studies. This may be best understood by considering them in the context of both quantitative (statistical) hypotheses and qualitative or conceptual hypotheses (maintained, diagnostic, and remedial hypotheses) introduced in chapter 3 (Tweeten, 1983). It is not uncommon for even quantitative hypotheses to be left implicit or not formally stated. Consider that the hypotheses in the proposal in appendix A are stated implicitly—as expected outcomes of the policies examined. There is no explicit hypothesis in the proposal in appendix C. In the example of the hypothesis about wine consumption above, researchers often forego the formal statement of the hypothesis regarding the expected relationships. We also often choose not to state the formal hypotheses about expected signs or magnitudes of the parameters in the relationship (e.g., to hypothesize an inverse relationship between wine price and wine consumption, *ceteris paribus*, or a direct relationship between prices for other beverages and wine consumption, etc.).

There is justification for not stating some hypotheses explicitly, particularly in certain types of research reporting (e.g., disciplinary journal articles) when the hypotheses are already implied. Many economists follow this practice with journal writing and even with research proposals for some audiences. However, we should be cautious with the practice because it carries the risk of compromising the rigor in our analytical thinking. For students developing research proposals for their advisors and advisory committees, a safe guide is that all hypotheses be made explicit; it lends rigor to the process.

Quantitative hypotheses are subject to empirical testing, the most common form being a statistical test. As such, we can identify four general characteristics of hypotheses that facilitate empirical testing (Williams, 1984):

- A. *They must be stated in specific terms.* Vague or general statements do not lend themselves to empirical testing.
- B. *They require that appropriate data be available or can be generated.* Empirical testing requires quantitative data.
- C. *They must be framed so that analytical techniques are available.*
- D. *They must have a conceptual basis.* Without the conceptual reasoning, quantification constitutes only association, not causation.

On the matter of qualitative hypotheses in the conceptual framework, it is common for all three types—maintained, diagnostic, and remedial hypotheses—to be used in the conceptual analysis of the problem. Maintained hypotheses often take the form of assumptions we make regarding the conditions in which the problem exists or the relevant dimensions of the problem. When analyzing issues or problems in the commodity production segment of the agricultural sector, for example, we often assume that the segment is competitive (farmers are price takers in both input and product markets, widespread availability of information, etc.), at least in regard to commodities such as grains, soybeans, and cotton. This assumption constitutes a maintained hypothesis. There are no formal quantitative tests for its acceptability, but the “test” is instead by general agreement or disagreement based on logic. Some who cannot accept your assumptions about the framework of the problem as you present it may not accept your analysis of it, thereby rejecting your proposed research altogether. However, some who cannot accept your assumptions will accept your proposed research if the assumptions are defensible.

A point that is especially relevant regarding maintained hypotheses is that the validity of an assumption, and the defensibility of it, often lies in the framing of the problem itself. Consider a research problem that focuses on understanding the forces determining the general level of cotton prices in the United States. We know that in a strict sense the market is not perfectly competitive. Among other things, cotton is not a homogeneous product (it differs in quality among countries, regions within the United States, and even individual bales produced by the same farmer). Yet for a given level of market aggregation, one may treat the market as competitive (use the perfectly competitive market model) for the purposes at hand. This maintained hypothesis (assumption) may be justifiable because the quality differences among countries and regions are stable, the problem involves only understanding the general level of price, and the general price is understood to be only an indicator of what is really a distribution of prices.¹

Diagnostic hypotheses—propositions about the causes of a problem—have an obvious place in the conceptual framework. The diagnostic hypotheses embodied in the conceptual frameworks of the proposals in appendices A and C are examples. In some instances the conceptual analysis of the problem may begin with a diagnostic hypothesis of the cause of the problem, then proceed to support that hypothesis with theoretical constructions. In this type of case, the hypothesis is either supported by the logical construction of theoretical reasoning or it is not. In other instances, the conceptual analysis of the perceived problem may lead the researcher to a diagnostic hypothesis about the cause(s) of the problem. The theoretical reasoning is essentially the same in both instances, and the test of logical reasoning is the same. The reader, in fact, may not be able to distinguish the researcher's point of shifting from diagnosis of causation to the diagnosis of potential solutions.

Consider, for instance, a research problem concerned with a country's chronic trade deficit. The researcher might present the diagnostic hypothesis that

its causes are in part due to monetary policy, in part to fiscal policy, in part to a policy on foreign investment within the country, and in part to domestic food policies that place ceilings on commodity prices within the country. The conceptual framework would, in its theoretical analysis, show why or how each of these policies contributes to the country's trade deficit. (Note that this analysis also would embody maintained hypotheses.)

Remedial hypotheses may be included in the conceptual framework, but are best viewed as optional. As noted in chapter 3, they are closely related to diagnostic hypotheses. Diagnostic hypotheses address the causes of the problem, while remedial hypotheses address solutions.² Maintained and diagnostic hypotheses can be tested only by logic of the reasoning as presented. Remedial hypotheses may be evaluated by logic of the reasoning, but they may be presented in such a way as to be empirically testable. Using the previous trade deficit problem example, a hypothesis of the form that a specified combination of monetary, fiscal, investment, and agricultural price policies would reduce or eliminate the trade deficit constitutes a remedial hypothesis. While this hypothesis is testable by weight of logic, it also may be empirically testable via developing an econometric model that captures the effects of these various forces on the country's international trade, gathering data and estimating model parameters, then simulating selected policy change scenarios with it. This approach still relies on logical constructions (of the model), but expected outcomes are empirical (quantified).

Whether hypotheses are quantitative or qualitative, they are an integral part of conceptualizing and the analytical research process. Goode and Hatt (1952, p. 56) state:

The formulation of the deduction constitutes a hypothesis; if verified, it becomes a part of a future theoretical construction. It is thus clear that the relation between the hypothesis and theory is very close indeed. While it is true that the two can never be satisfactorily separated, it is useful to think of them as two aspects of the way in which science adds to knowledge. Thus a theory states a logical relationship between facts. From this theory other propositions [hypotheses] can be deduced that should be true if the first relationship holds. A hypothesis looks forward. It is a proposition which can be put to a test to determine its validity.

Goode and Hatt proceed to identify three main difficulties people have in developing hypotheses (p. 57). The first is an absence of a clear theoretical framework, or absence of knowledge of theory. The second is the inability to use the framework logically—to see the hypotheses within it. Both difficulties might arise from an insufficient understanding of the body of theory or inability to integrate it into one's reasoning process (these may amount to the same thing). In any event, both difficulties are encountered in both quantitative and qualitative hypotheses. The third difficulty, failure to understand research techniques that allow them to be tested, applies more specifically to quantitative hypotheses.

Ladd (1991) would add a fourth difficulty: that of relating the theoretical hypotheses (in terms of ideal types) to the real types of the research problem.

OPERATIONAL SUGGESTIONS

As you begin to develop your conceptual framework, keep a clear focus on its primary purpose—to analyze the research problem. Its secondary or subsidiary functions will be facilitated by maintaining your attention on the main point of the exercise. Also, keep in mind that the conceptualizing process involves abstractions and abstract reasoning, but applied to the concrete concerns of the issues surrounding the problem(s) when the research is subject-matter or problem-solving focused. Your conceptual analysis will necessarily involve abstractions (simplifications), but with a definite focus—analyzing the problem. Larrabee (1964, p. 171) discusses a “happy medium” that is relevant:

The happy medium is a constant, cautious shuttling back and forth between words and things. No one can afford to let either one monopolize his time and attention. The “highbrow” is prone to overdo abstractions and to neglect the facts; while the “lowbrow” is just as likely to mock at the theories which he does not understand. To condemn all abstraction and all ambiguity, just because they are subject to certain vices when used in particular ways, is just as silly as to cover them with a blanket of eulogy for their virtues in other contexts. We must never forget that thinking is an art as well as a science; and that different languages may be needed for different purposes. Our policy in regard to abstraction, then, should not be inflexible, but adapted to our purposes.

As you approach the process of conceptualizing, then organizing, a method that is often helpful is to start with simple conceptual models and build on them to the degree of complexity and sophistication that is necessary for the task. This general approach to conceptualizing can be both productive and efficient, irrespective of the target of the conceptualizations. That is, it is useful for conceptualizing when involved in research, policy analysis, and assisting business people, administrators, the public, and others. One must recognize that it is first necessary to define the problem sufficiently to apply some analysis to it. More specifically, there are some steps that may be useful in applying this general guideline:

A. Most importantly, *gain a grasp of the research literature related to your problem*. How have others conceptualized problems similar to yours? How much, or what parts, of those conceptual analyses are adaptable to your conceptual framework? Be aware, however, that you are not likely to find other conceptual frameworks that fit your problem exactly. Modifications will be needed even when similar conceptualizations are available in the literature. You need not reinvent the wheel, but you may need to impose some design modifications to make it fit your wagon.

B. *Reduce the problem to the simplest, most uncluttered set of conditions possible* as a point of departure. Get the problem confined to its most basic dimensions. This is done because the human brain cannot manipulate large numbers of variables or interacting ideas simultaneously (Williams, 1984). You may use assumptions to ignore aspects of the problem that you believe to have little or no bearing on the forces causing it or that might alter its nature. You may even want initially to assume away parts of the problem so as to make it simpler to gain understanding of its dimensions. In either case, these simplifying assumptions (maintained hypotheses) should undergo periodic reexamination throughout the research process (more on this in item D below).

C. *Identify the applicable economic theory.* If the research problem has an international or interregional trade component, then trade theory probably will be applicable. There also could be, for example, reasons for including certain aspects of production theory, welfare theory, and so on. One also may want to identify and apply alternative theoretical formulations to the problem (Castle, 1989, pp. 5-6). There is a range of alternate theoretical formulations from which to draw and little reason to confine oneself to any particular subset or to a doctrine "school of thought."

D. *Start with a "base model" analysis.* Using the carefully constructed simplifying assumptions from step A, proceed to construct a "simple" conceptual economic model of the causes of the problem. This may be, for example, a short-run, comparative statics approach to an analysis. It may employ some graphical analysis in the process, although it could proceed on to some more multidimensional or multivariable mathematical analysis. The primary purpose of this step is to get a solid logical beginning on the conceptual analysis. It is unlikely, however, that the conceptual analysis will be complete with this single step, particularly with more sophisticated economic analyses. Most individual research problems require working through multiple successive iterations of conceptualizing before developing something that does justice to understanding the various dimensions of the problem.

E. *Expand the base model analysis to other relevant dimensions of the problem.* Alter and relax restricting assumptions, then reanalyze the dimensions of the problem under modified assumptions. This could, for example, bring in long-run as well as short-run analyses, introduce dynamic aspects to augment the comparative statics, and generally make the conceptualization more complex. Note, however, that the goal is not complexity; it is to get as close to the relevant dimensions of the problem as conditions will permit. There are circumstances when added complexity serves to confuse the issues rather than clarify them.

F. *Assemble relevant, testable hypotheses from the conceptual analysis.* You will have been employing some maintained and diagnostic hypotheses in the process of conducting the conceptual analysis. If your analysis has captured the important dimensions of the problem, adapted the relevant economic theory to analyzing it, and analyzed it effectively, then those expected outcomes should

provide a set of testable hypotheses. These may be stated in more general form than formal statistical hypotheses, and they may need to be arranged in some logical order.

There are abundant examples of conceptual frameworks. Economics journals are replete with examples, although they are not always labeled as such; a conceptual framework may be identified with a heading such as model (not to be confused with an empirical model), conceptual model, conceptual considerations, theoretical considerations, or it may simply be integrated into the problem description. In some instances a journal article consists entirely of a conceptual framework (see Smith [1968] for an example). However, the norm for the conceptual framework in a graduate thesis or dissertation is for it to be presented in a separate, formal chapter or section.

The same guidelines for writing covered in previous chapters apply to writing the conceptual framework. Anticipate the writing of the conceptual framework to be rigorous. Expect to write multiple drafts. Seek critical reviews of drafts from persons with demonstrated ability in conceptualizing. Recognize the conceptual framework as a creative and imaginative endeavor (chapter 2) rather than a technical adaptation of theory. Locate and study well-constructed conceptual frameworks.

SUMMARY

The conceptual framework is an often misunderstood component of a research proposal. Its central purpose is to conduct a conceptual analysis of the research problem. It is fundamental to understanding and explaining the research problem. It is not less important in subject-matter and problem-solving research than in disciplinary research. Confusing this primary role with some of its secondary benefits—a theoretical link between objectives and methods and procedures and a foundation for empirical models, for example—is a common problem in constructing conceptual frameworks.

Existing economic theory is the capital stock from which conceptual frameworks are derived, but generalized theory does not, by itself, comprise a conceptual framework (except in the cases of purely theoretical disciplinary research). The conceptual framework entails adaptation of general theories to the specific problem and analysis of the problem using the theory. The adaptation process may require theoretical refinements or developing new theoretical constructions. Most theoretical advancements occur through the need to analyze or explain a problem.

Hypothesis formulation and testing is intertwined with the process of conceptualizing. We are generally familiar with conceptualization resulting in empirically testable (quantitative) hypotheses. Qualitative hypotheses—maintained,

diagnostic, and remedial—are also integral components of the conceptual framework, albeit less commonly recognized explicitly. These matters are discussed, along with conditions for quantitative hypotheses.

Some procedural guidelines for formulating and writing the conceptual framework in a research proposal are provided. These include a thorough grounding in the theoretical and other literature related to the problem, identifying relevant economic theory, initiating conceptual analysis with simple formulations and building needed complexity and sophistication on them, and formulating hypotheses directly from the conceptual results.

SUGGESTED EXERCISES

1. Choose and read an article from a recent economics journal. Does it have a distinctly separate conceptual framework (it need not be labeled “Conceptual Framework”)? If it does, describe and evaluate it; does it accomplish its intended purpose? If it is not explicit, are elements of the conceptual framework integrated into other parts of the article? Explain. If there is no conceptual framework within the article, explicit or implicit, outline your suggested conceptual framework for the paper.

2. Select a graduate thesis or dissertation and conduct the same analysis of it as for the journal article in exercise 1.

3. Using the two sources from exercises 1 and 2, identify and briefly explain the hypotheses in each. Are the hypotheses explicit or implicit? Address yourself to both quantitative and qualitative hypotheses. Are the hypotheses tested within the article and/or thesis or dissertation?

NOTES

1. While the assumption of a competitive market structure is used to illustrate a maintained hypothesis, competitive markets as a maintained hypothesis is, in the author's opinion, overused by economists, especially in applied empirical estimation research. It appears to be frequently adopted because it facilitates construction of empirical models, not because it has been conceptually defended and justified. Any maintained hypothesis made solely for empirical convenience is likely to result in unreliable research results and misinformation.

2. It may be noted that both remedial and diagnostic hypotheses may also be included in the conclusions of a research project. The conclusions, discussed in chapter 10, may lead one to offer hypotheses regarding the causes or solutions to a problem.